DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning

INTERNSHIP REPORT

ON

"DATA ANALYST"

Submitted in partial fulfillment for the award of degree(21AI8ICINT)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

Krutika R Sinnur - 1DS21AI028

Conducted at

CodTech IT Solutions Pvt.Ltd
IT Services and IT Consulting

Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-560078

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CERTIFICATE

This is to certify that the Internship titled **"Data Analyst"** carried out by **Ms Krutika R Sinnur [1DS21AI028]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2024-2025. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (21AI8ICINT)

N. Santhosh

Signature of Reporting Head	Signature of Internship Coordinator	Signature of HoD
Neela Santhosh Kumar Human Resources & Academic Head CodTech IT Solutions	**Dr. Vindhya P Malagi** Dept. of AI & ML DSCE	**Dr. Vindhya P Malagi** Dept. of AI & ML DSCE

External Viva:

Name of the Examiner	Signature with Date
1) ________________	________________
2) ________________	________________

DAYANANDA SAGAR COLLEGE OF ENGINEERING

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Department of Artificial Intelligence and Machine Learning

DECLARATION

I, **Krutika R Sinnur,** final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in **CodTech IT Solutions Pvt.Ltd**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor's degree in Artificial Intelligence and Machine Learning, during the academic year 2024- 2025.

Date: USN: 1DS21AI028

Place: Bangalore NAME: KRUTIKA R SINNUR

OFFER LETTER

CODTECH IT SOLUTIONS PVT.LTD

IT SERVICES & IT CONSULTING

8-7-7/2, Plot NO.51, Opp: Naveena School, Hasthinapuram Central, Hyderabad , 500 079. Telangana

Internship Offer Letter

Date : 18/02/2025

Intern ID :CT4MVMX

Dear Krutika R Sinnur

Congratulations on being selected for the **Data Analytics.** We at **CODTECH IT SOLUTIONS PVT.LTD** are thrilled to have you join our team. **This Online Internship will span 4 months from February 20th, 2025 to June 20th, 2025.**

This internship is designed as an educational experience, focusing on learning, skill development, and gaining practical knowledge. As an intern, we expect you to:

1. Complete all assignments to the best of your ability.
2. Follow any lawful and reasonable instructions provided by your supervisors.
3. Participate actively in team meetings and discussions.
4. Provide regular updates on your progress.
5. Adhere to company policies and maintain a professional demeanor.
6. Collaborate effectively with team members and contribute to group projects.
7. Seek feedback and apply it to improve your performance.

We trust that you will approach all tasks with diligence and enthusiasm. We are confident that this internship will be an enriching experience for you. We look forward to working with you and supporting you in achieving your career aspirations

Best regards,

N. Santhosh

Neela Santhosh Kumar

Human Resources & Academic Head

CODTECH IT SOLUTIONS PRIVATE LIMITED

VERIFIED BY

CODTECH IT SOLUTIONS PVT LTD

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COMPLETION CERTIFICATE

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

I express my sincere thanks to our Principal **Dr. B G Prasad**, for providing us adequate facilities to undertake this Internship.

I would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

I would like to thank, **Neela Santhosh Kumar** Human Resources & Academic Head, Codtech IT Solutions Private Limited for guiding us during the period of internship.

I express my deep and profound gratitude to our Internship Co-ordinator, **Dr.Vindhya P Malagi,** Head Of Department – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

I would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

I would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, I would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

Krutika R Sinnur [1DS21AI028]

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CHAPTER 1
INTRODUCTION

The rapid growth of data and its increasing importance across industries has made data analysis a critical skill in today's technological landscape. In this context, an internship was undertaken at CodTech IT Solutions Pvt. Ltd, a company recognized for delivering advanced IT services and consulting solutions. This internship offered a platform to work with real-time datasets, implement data processing techniques, and develop predictive models and dashboards, effectively bridging the gap between theoretical knowledge and practical application.

The internship was organized into four major tasks, each focusing on a specific area within the domain of data science and analytics. These tasks were thoughtfully crafted to simulate real-world challenges encountered by data professionals, providing exposure to various tools and technologies used across the industry. The assignments encompassed areas such as big data analysis using scalable tools like PySpark or Dask, predictive modeling through machine learning techniques, interactive dashboard creation with data visualization tools, and sentiment analysis using natural language processing (NLP).

The primary objective of the internship was to build the capability to extract meaningful insights from complex datasets and to present those insights in a clear and actionable format. Independent work on each task contributed to the development of strong problem-solving abilities, an understanding of the end-to-end data pipeline, and proficiency in widely used tools such as Python, pandas, matplotlib, scikit-learn, Tableau, and NLP libraries. The structure of the internship encouraged self-learning through the use of online resources including YouTube tutorials, ChatGPT, and official documentation. This approach not only strengthened technical competencies but also fostered independence and adaptability—key attributes in a dynamic technological environment.

In conclusion, the internship at CodTech IT Solutions Pvt. Ltd provided a comprehensive and valuable experience in data analytics. It served as a pivotal step in transitioning from academic learning to real-world application. The tasks completed during this period are detailed in the following chapters, highlighting the skills acquired, challenges encountered, and knowledge gained throughout the internship journey.

CHAPTER 2
DESCRIPTION OF ORGANIZATION

CODTECH IT SOLUTIONS PVT. LTD

ABOUT THE COMAPNY

CodTech IT Solutions Pvt. Ltd is a growing player in the IT Services and Consulting domain, focused on delivering comprehensive digital transformation solutions to businesses of all sizes. The company specializes in offering services across data analytics, software development, cloud computing, artificial intelligence, and IT consulting. With a mission to empower organizations through innovative technology, CodTech helps its clients improve efficiency, gain insights from their data, and stay competitive in a rapidly evolving digital ecosystem.

CodTech places a strong emphasis on skill development and learning, and it actively engages with students and young professionals through internship programs, hands-on projects, and guided training sessions. These initiatives are designed to bridge the gap between academic knowledge and industry expectations, making it a preferred destination for students aspiring to build careers in tech. The company fosters a collaborative and inclusive environment, encouraging interns to explore real-world tools, solve practical problems, and become proficient in core IT competencies. Interns are supported through task-based learning, clear deliverables, and access to self-learning resources, making CodTech a valuable launchpad for future tech professionals.

VISION:

To be a trusted global leader in providing cutting-edge IT solutions that drive innovation, efficiency, and value for clients, while nurturing the next generation of technology professionals.

MISSION:

- To deliver scalable, secure, and intelligent IT solutions that address real business challenges.
- To empower learners and young professionals through practical, task-based training and internship programs.

- To continuously innovate and adopt the latest technologies in data analytics, software development, and artificial intelligence.
- To create a culture of learning, collaboration, and excellence that benefits both clients and aspiring technologists.

CORE AREAS OF INNOVATION:

At CodTech IT Solutions Pvt. Ltd, innovation is at the heart of every service and solution offered. The company is committed to staying at the forefront of technological advancements by continuously exploring and integrating new-age tools and methodologies. Its core areas of innovation include:

1. Data Analytics and Business Intelligence

CodTech focuses on transforming raw data into actionable insights through advanced analytics techniques. The company explores big data technologies like PySpark and Dask for scalable processing, and uses tools like Tableau and Power BI to develop interactive dashboards that help clients make informed decisions.

2. Machine Learning and Artificial Intelligence

Innovation in predictive modeling, classification, clustering, and recommendation systems forms a key part of CodTech’s offerings. By using supervised and unsupervised learning algorithms, the company builds intelligent systems capable of learning from data to automate complex decision-making processes.

3. Natural Language Processing (NLP)

With a growing emphasis on customer experience and social media analytics, CodTech has invested in NLP technologies for sentiment analysis, emotion detection, and language modeling. These innovations allow businesses to better understand user opinions, behaviors, and trends.

4. Cloud Computing and Scalable Solutions

CodTech actively integrates cloud technologies to offer secure, scalable, and cost-effective IT solutions. These cloud services enhance data storage, model deployment, and cross-platform accessibility for clients operating in dynamic environments.

5. Education and Skill Development Platforms

Beyond client services, CodTech also innovates in the learning and development space. By providing guided internships, structured tasks, and access to real-world tools, the company has created a unique learning ecosystem that promotes hands-on skill acquisition for aspiring technologists.

These core areas of innovation not only enhance CodTech's ability to deliver value-driven solutions to clients but also reflect its commitment to continuous learning, improvement, and thought leadership in the IT services industry.

CHAPTER 3
SCOPE OF WORK

The internship at **CodTech IT Solutions Pvt. Ltd** was designed to offer a comprehensive, task-driven learning experience, combining theoretical understanding with practical application. The scope of work encompassed four distinct yet interrelated projects, each targeting a fundamental area within the field of data science and analytics. These tasks were strategically structured to enhance both technical skills and problem-solving capabilities while encouraging self-directed learning and efficient project execution.

Each task addressed real-world challenges and required the application of contemporary tools and techniques, such as PySpark for scalable data processing, machine learning for predictive modeling, Tableau/Power BI for interactive dashboards, and natural language processing (NLP) libraries for sentiment analysis. The practical implementation of each module deepened my understanding of how data can be leveraged for insightful, actionable outcomes in business and technology contexts.

1. Big Data Analysis using PySpark

This task emphasized the need for handling and analyzing large datasets using distributed computing frameworks. Traditional data processing libraries like pandas struggle with large-scale data due to memory limitations. PySpark provide the ability to process massive datasets in parallel, making them suitable for big data environments.

Key components of this task included:

- Loading and inspecting large datasets from external sources.
- Performing complex transformations using PySpark DataFrames, such as filtering based on conditions, grouping data, and computing aggregations like averages and counts.
- Joining multiple datasets efficiently to uncover relationships.
- Deriving meaningful insights such as customer behavior patterns, product trends, or performance metrics.
- Writing optimized scripts to ensure scalability and minimize processing time.

This task helped in gaining experience in big data workflows and learned how to write code that is both scalable and efficient—an essential skill in enterprise-level data engineering roles.

2. Predictive Analysis using Machine Learning

The second task focused on building a machine learning model to predict future outcomes based on historical data. This involved the complete machine learning pipeline, starting from raw data to model deployment.

Steps undertaken in this task:

- Data exploration and cleaning, addressing issues such as missing values, duplicate records, and inconsistent formatting.
- Feature selection and engineering, identifying relevant predictors that contribute to the outcome and creating new derived variables.
- Splitting the dataset into training and test sets to validate the model's performance.
- Training multiple models, such as logistic regression, decision trees, or support vector machines, using scikit-learn.
- Evaluating performance metrics like accuracy, confusion matrix, ROC-AUC score, and precision/recall to choose the best-performing model.
- Tuning hyperparameters using grid search or cross-validation to improve model generalization.

3. Interactive Dashboard Development

Data visualization is crucial in transforming complex data into easily understandable formats for non-technical stakeholders. In this task, an interactive dashboard was developed using tools such as Tableau, Power BI,

Key highlights of this task included:

- Cleaning and structuring the data in a format suitable for dashboard integration.
- Designing charts and KPIs that highlight important metrics such as trends, growth rates, and category-wise comparisons.
- Creating interactive filters, dropdowns, and slicers for user customization.
- Using calculated fields and parameters to enhance dashboard functionality.
- Focusing on usability and visual appeal, following best practices in data storytelling.

4. Sentiment Analysis using Natural Language Processing (NLP)

In the final task, the focus was on text analytics, specifically performing sentiment analysis on unstructured textual data such as tweets, product reviews, or feedback forms. This task proved to be one of the most impactful, as it combined programming, linguistics, and machine learning.

Key steps included:

- Text preprocessing: Removing stop words, punctuation, and noise; tokenizing and normalizing text using libraries like NLTK and spaCy.
- Vectorization: Transforming textual data into numerical representations using TF-IDF or word embeddings like Word2Vec.
- Model training: Building classifiers such as Naive Bayes or LSTM (Long Short-Term Memory) networks for sentiment detection (positive, neutral, negative).
- Evaluating performance: Using accuracy, F1-score, and confusion matrices to assess model effectiveness.
- Visualizing sentiment trends across categories or time periods to derive insights about public opinion or customer satisfaction.

CHAPTER 4
LEARNING OBJECTIVES

The internship at CodTech IT Solutions Pvt. Ltd was carefully structured to serve as a platform for applied learning, focusing on building industry-relevant skills in data analysis, machine learning, data visualization, and natural language processing (NLP). The program was designed not only to strengthen technical expertise but also to instil analytical thinking, problem-solving capabilities, and the ability to independently manage and execute data-driven projects.

The following are the detailed learning objectives that guided the structure and execution of the internship:

1. **Proficiency in Big Data Processing and Distributed Computing**

One of the core learning goals was to understand how to work with large-scale datasets that exceed the memory limits of traditional tools. This objective emphasized:

- Gaining familiarity with PySpark, a powerful framework that enable parallel processing and distributed computing.
- Learning how to efficiently load, transform, and manipulate big data using DataFrame APIs and lazy execution mechanisms.
- Applying aggregation, filtering, and join operations in a scalable manner that mimics enterprise-level data engineering tasks.
- Developing the ability to write performance-optimized code suitable for high-volume data analysis.

This helped build a foundation for handling large, complex datasets in environments where speed and scalability are essential.

2. **Application of Supervised Machine Learning Algorithms**

Another important objective was to apply machine learning techniques to solve classification and regression problems. This included:

- Understanding the machine learning pipeline, including stages such as data cleaning, feature engineering, model training, validation, and tuning.
- Exploring the use of algorithms such as Logistic Regression, Decision Trees, and Support Vector Machines to build predictive models.
- Applying performance metrics like accuracy, precision, recall, confusion matrix, and ROC-AUC to evaluate model outcomes.

- Understanding the concept of overfitting and underfitting, and implementing techniques like regularization and cross-validation to improve generalization.

This part of the internship was crucial for understanding how predictive models are developed and validated in real-world scenarios.

3. Designing and Building Interactive Dashboards

Communicating data effectively through visualization was another major focus of the internship. This objective aimed to:

- Introduce the use of professional dashboarding tools such as Tableau, Power BI, or Dash.
- Teach best practices in data storytelling, including how to select appropriate chart types, design layouts, and highlight key performance indicators (KPIs).
- Emphasize user experience (UX) in dashboard design through the use of slicers, filters, and interactive elements.
- Train on real-time data connectivity and dashboard responsiveness for executive-level reporting.

This component strengthened the ability to deliver business insights in a clear and interactive format suitable for non-technical audiences.

4. Implementing Sentiment Analysis using Natural Language Processing

The internship also covered advanced topics in Natural Language Processing (NLP), with a specific focus on sentiment analysis. The objective was to:

- Learn text preprocessing techniques, including tokenization, stopword removal, stemming, lemmatization, and normalization.
- Convert text into numerical vectors using TF-IDF, CountVectorizer, and word embeddings.
- Build and evaluate sentiment classification models using Naive Bayes, Logistic Regression, and LSTM networks.
- Analyze emotion trends from textual data such as product reviews, social media posts, or survey feedback.

This enabled the development of intelligent systems capable of interpreting human language, a critical skill in today’s data-centric industries.

5. Familiarity with Industry Tools and Programming Environments

To support all technical objectives, the internship also aimed at building fluency in industry-standard tools and environments, including:

- Python programming with libraries such as pandas, NumPy, matplotlib, seaborn, scikit-learn, NLTK, and spaCy.
- Version control and collaboration using GitHub repositories for storing and documenting code.
- Integrated development environments (IDEs) such as Jupyter Notebook, VS Code, and Google Colab for interactive coding and sharing outputs.

This allowed for a professional-level approach to coding, documentation, and reproducibility.

6. Development of Analytical and Professional Skills

In addition to technical competencies, the internship was also geared toward enhancing soft skills and professional attributes:

- Cultivating analytical thinking and decision-making through independent problem-solving.
- Practicing effective time management and task prioritization while working under deadlines.
- Building technical writing and presentation skills through structured documentation and reporting of outcomes.
- Encouraging self-learning by making use of online tutorials, documentation, and forums to research solutions.

The learning objectives of the internship collectively contributed to a well-rounded, hands-on experience that aligned with the expectations of industry roles in data science, analytics, and machine learning. These objectives laid a strong foundation for future work in the domain, enabling the application of learned skills in real-time business environments.

CHAPTER 5

CHALLENGES AND SOULTION

The internship experience at **CodTech IT Solutions Pvt. Ltd** was both intellectually stimulating and technically demanding. Each task presented a distinct set of challenges that mirrored the complexities faced in real-world data projects. These challenges spanned various stages of the data analysis pipeline—from data acquisition and processing to model deployment and visualization. Addressing them required a combination of technical strategies, tool optimization, and effective problem-solving methodologies.

Outlined below are the key challenges encountered during the internship, along with the corresponding solutions that were implemented to ensure task completion and learning outcomes.

1. **Managing Large and Complex Datasets**

   <u>Challenge:</u>

   One of the initial hurdles involved handling large-scale datasets during the big data analysis task. Traditional data analysis libraries like pandas were not capable of efficiently managing datasets that exceeded system memory, leading to significant lags or crashes.

   <u>Solution:</u>

   To address scalability, distributed computing frameworks such as PySpark was employed. These tools allowed for partitioning datasets into manageable chunks and performing parallel processing across multiple cores or clusters. Optimization techniques such as lazy evaluation, caching, and broadcast joins were also used to enhance efficiency and minimize memory usage. This shift enabled smooth processing of high-volume data while maintaining performance.

2. **Ensuring Data Quality and Consistency**

   <u>Challenge:</u>

   Datasets collected for analysis often contained inconsistencies such as missing values, duplicate records, irrelevant features, and non-standard formats. These quality issues affected the reliability of both analytical outputs and machine learning models.

   <u>Solution:</u>

   Comprehensive data cleaning and preprocessing steps were applied to standardize and prepare the data. Techniques included:

- Imputation of missing values using statistical methods.
- Removal of duplicates and irrelevant columns.
- Normalization and encoding of categorical data.
- Outlier detection to eliminate anomalies.

Automated pipelines were built using Python libraries like pandas, NumPy, and scikit-learn, ensuring that data fed into models and dashboards was clean, consistent, and analysis-ready.

3. Feature Selection and Model Overfitting

Challenge:

In the machine learning task, selecting appropriate features without overfitting the model posed a significant challenge. Inclusion of irrelevant or redundant features often led to high training accuracy but poor performance on unseen data.

Solution:

Feature selection techniques such as correlation matrix analysis, recursive feature elimination (RFE), and mutual information scores were used to identify the most predictive attributes. Models were trained using cross-validation, and regularization techniques like Ridge and Lasso regression were applied to prevent overfitting. These practices ensured the creation of models that were both accurate and generalizable.

4. Designing Intuitive and Responsive Dashboards

Challenge:

While developing interactive dashboards, challenges arose in balancing design aesthetics with functionality. Initial drafts sometimes included cluttered visuals, improper chart types, or low usability for end users.

Solution:

To improve dashboard design, principles of data storytelling and user-centered design were adopted. The following best practices were followed:

- Use of appropriate chart types to match the data context (e.g., bar charts for comparisons, line charts for trends).
- Implementation of filters and slicers for interactivity.
- Structuring dashboards into logical sections with clear headings and KPIs.
- Maintaining color consistency and readable fonts for accessibility.

5. Low Accuracy in Sentiment Classification Models

Challenge:

The NLP task presented challenges in sentiment classification accuracy due to the nature of textual data. Issues such as sarcasm, ambiguous phrases, and slang created noise in the training data, leading to sub-optimal model performance.

Solution:

A multi-step approach was taken to improve model accuracy:

- Advanced text preprocessing was implemented using libraries like NLTK and spaCy, including lemmatization, punctuation removal, and named entity recognition.
- Various vectorization techniques were tested, including TF-IDF, CountVectorizer, and pre-trained embeddings like GloVe.
- Multiple models were evaluated (Naive Bayes, Logistic Regression, and LSTM) to determine the most suitable approach.
- Hyperparameter tuning and model stacking were explored to improve generalization.

The outcome was a sentiment classification model capable of producing more accurate predictions, even on nuanced textual inputs.

6. Ensuring Code Readability and Reusability

Challenge:

Maintaining code that was modular, readable, and reusable was essential for documentation and future use. However, initial scripts lacked structure and documentation.

Solution:

Coding best practices were enforced across all tasks, including:

- Writing modular functions and reusable components.
- Adding inline comments and docstrings for clarity.
- Using GitHub repositories for version control and collaborative tracking.

This improved the maintainability of the codebase and aligned the workflow with industry development standards.

HARDWARE AND SOFTWARE REQUIREMENTS

Hardware requirements:

Component	Specification
Processor (CPU)	Minimum: Intel Core i5 / AMD Ryzen 5, Recommended: Intel Core i7 / Ryzen 7 or higher
RAM	Minimum: 8 GB, Recommended: 16 GB or higher
Storage	Minimum: 256 GB SSDRecommended: 512 GB SSD or higher
Graphics	Optional: Dedicated GPU (NVIDIA GTX/RTX) for deep learning models
Internet Connection	Stable broadband or Wi-Fi connection for data access, tool installation, and cloud operations

Software Requirements:

Category	Tools/Technologies Used
Operating System	Windows 10/11
Programming Language	Python 3.7 or above
Development Tools	Jupyter Notebook, Google Colab, Visual Studio Code (VS Code)
Libraries & Frameworks	pandas, NumPy, matplotlib, seaborn, scikit-learn, NLTK, spaCy, PySpark
Big Data Tools	PySpark (local installation or Colab environment)
Visualization Tools	Tableau Public, Power BI Desktop
Version Control	Git, GitHub
NLP Utilities	HuggingFace Transformers, NLTK, spaCy
Browser	Google Chrome, Microsoft Edge, or Firefox for dashboard access and web-based notebooks

CHAPTER 6

ACHEIVEMENTS AND CONTRIBUTION

The internship at **CodTech IT Solutions Pvt. Ltd** resulted in several significant contributions across multiple technical domains. Each task presented opportunities to apply conceptual knowledge in practical scenarios, leading to the successful development of data-driven solutions. The contributions made during the internship extended beyond task completion, as they also demonstrated creativity in problem-solving, consistency in performance, and commitment to producing high-quality outputs.

This chapter outlines the core achievements and technical contributions delivered during the internship tenure.

1. Task-Based Deliverables

Big Data Analysis (Task 1)

- Successfully implemented scalable data processing using **PySpark** , enabling real-time insights from large datasets.
- Delivered a well-structured **Python notebook/script** that handled high-volume data with optimized memory usage and runtime performance.
- Derived key insights through aggregations and group-based metrics such as trend identification, frequency distribution, and categorical comparisons.
- Ensured efficient data ingestion, transformation, and summarization pipelines suitable for enterprise-scale applications.

```
team_schema=StructType([
    StructField("team_sk", IntegerType(), True),
    StructField("team_id", IntegerType(), True),
    StructField("team_name", StringType(), True)
])
team_df=spark.read.schema(team_schema).format("csv").option("header","true").load("s3://ipl-data-analysi
```

```
team_df:  pyspark.sql.dataframe.DataFrame = [team_sk: integer, team_id: integer ... 1 more field]
```

Fig 6.1 – Loading Data into Spark DataFrame

```
#filter for include only valid deliveries
ball_by_ball_df=ball_by_ball_df.filter((col("wides")==0)& (col("noballs")==0))
#Aggregation cal the total and avg runs scored in each match ansd inning
total_and_avg_runs=ball_by_ball_df.groupBy("match_id","innings_no").agg(sum("runs_scored").alias("total_runs"),
avg("runs_scored").alias("avg_runs"))
```

Fig 6.2 - Data Cleaning and Aggregation Using PySpark

```
toss_impact_individual_matches = spark.sql("""
SELECT m.match_id, m.toss_winner, m.toss_name, m.match_winner,
       CASE WHEN m.toss_winner = m.match_winner THEN 'Won' ELSE 'Lost' END AS match_outcome
FROM match m
WHERE m.toss_name IS NOT NULL
ORDER BY m.match_id
""")
toss_impact_individual_matches.show()
```

```
toss_impact_individual_matches:  pyspark.sql.dataframe.DataFrame = [match_id: integer, toss_winner: string ... 3 more fields]

+--------+--------------------+---------+--------------------+-------------+
|match_id|         toss_winner|toss_name|        match_winner|match_outcome|
+--------+--------------------+---------+--------------------+-------------+
|  335987|Royal Challengers...|    field|Kolkata Knight Ri...|         Lost|
|  335988| Chennai Super Kings|      bat| Chennai Super Kings|          Won|
```

Fig 6.3 - Using SQL Queries in PySpark

Predictive Modeling (Task 2)

- Designed and developed a supervised machine learning pipeline for classification/regression tasks using scikit-learn.
- Built multiple models, performed feature selection, and applied **cross-validation** to improve prediction accuracy and model generalization.
- Delivered a complete **Google Colab notebook** with EDA, model evaluation, confusion matrix, and performance interpretation.
- Presented predictions with business-relevant insights and justifications for model selection.

```
# Check for missing values
print("Missing values:\n", df.isnull().sum())

# Distribution of quality scores
plt.figure(figsize=(8,4))
sns.countplot(x='quality', data=df)
plt.title("Wine Quality Score Distribution")
plt.show()

# Correlation heatmap
plt.figure(figsize=(10,8))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
plt.title("Feature Correlation Matrix")
plt.show()
```

Fig 6.4 – Exploratory Data Analysis code

```
# Convert quality to binary classification: 1 if quality >= 7, else 0
df['good_quality'] = (df['quality'] >= 7).astype(int)

# Features and target
X = df.drop(['quality', 'good_quality'], axis=1)
y = df['good_quality']
```

Fig 6.5 – Feature Engineering

```
# Select top 8 features
selector = SelectKBest(score_func=f_classif, k=8)
X_train_sel = selector.fit_transform(X_train_scaled, y_train)
X_test_sel = selector.transform(X_test_scaled)
```

Fig 6.6 – Feature Selection

```
# Train Random Forest
clf = RandomForestClassifier(n_estimators=100, random_state=42,
class_weight='balanced')
clf.fit(X_train_sel, y_train)
```

Fig 6.7 – Training the model

```
Classification Report:
              precision    recall  f1-score   support

           0       0.94      0.99      0.97       277
           1       0.93      0.60      0.73        43

    accuracy                           0.94       320
   macro avg       0.94      0.80      0.85       320
weighted avg       0.94      0.94      0.94       320
```

Fig 6.8 -Classification Report for the Wine Quality Classification Model

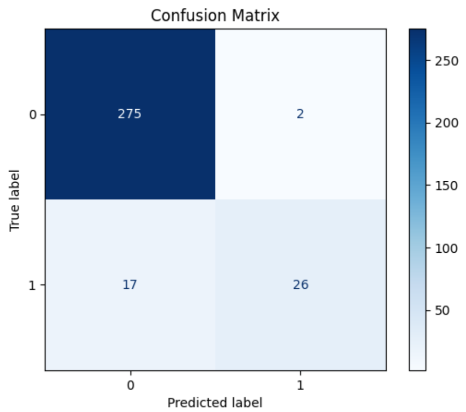

Fig 6.9 – Confusion Matrix for the Wine Quality Classification Model

Interactive Dashboard Development (Task 3)

- Developed a fully functional, **interactive dashboard** using tools like **Tableau** and **Power BI**, reflecting business insights through real-time data.
- Integrated multiple visual components (bar charts, pie charts, KPIs, filters) to ensure ease of navigation and clarity in insight delivery.
- Applied effective visual storytelling principles to make the dashboard accessible to both technical and non-technical users.
- Structured and presented datasets in a dashboard-ready format, improving efficiency in business decision-making processes.

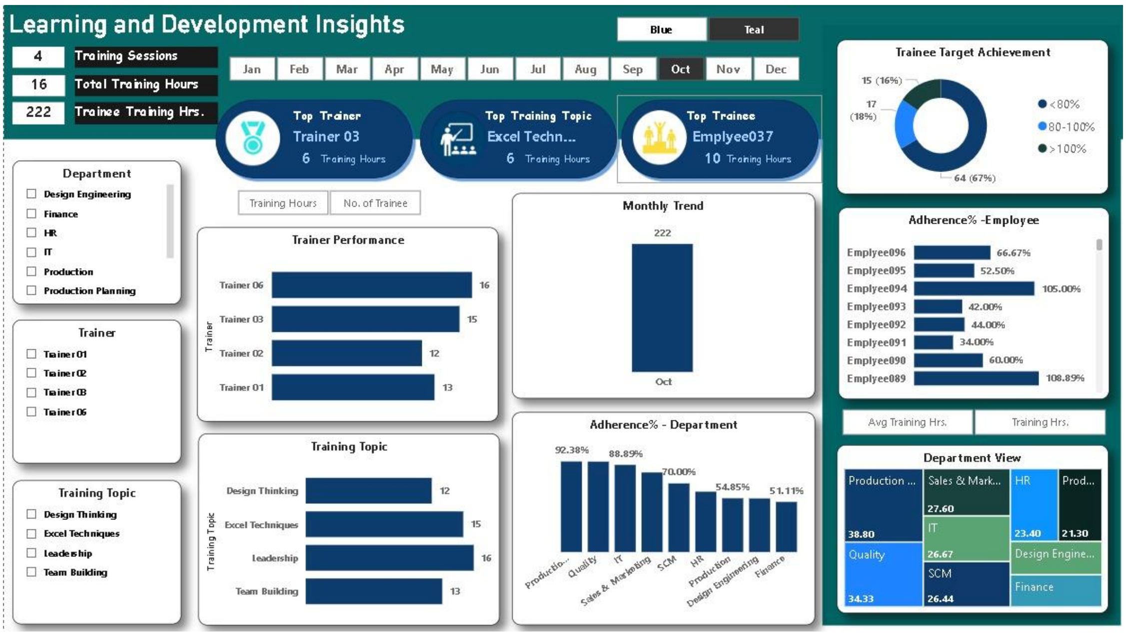

Fig 6.10 - Comprehensive Dashboard for Learning and Development Performance Metrics

Sentiment Analysis using NLP (Task 4)

- Implemented **sentiment classification models** using machine learning and NLP libraries such as NLTK, spaCy, and TF-IDF.
- Preprocessed and cleaned raw text data, vectorized input features, and trained models for classifying sentiments as positive, negative, or neutral.
- Evaluated the model using precision, recall, F1-score, and visualized sentiment trends for interpretability.
- Provided a comprehensive sentiment analysis pipeline that can be extended to real-world applications like product reviews, customer support, or social media monitoring.

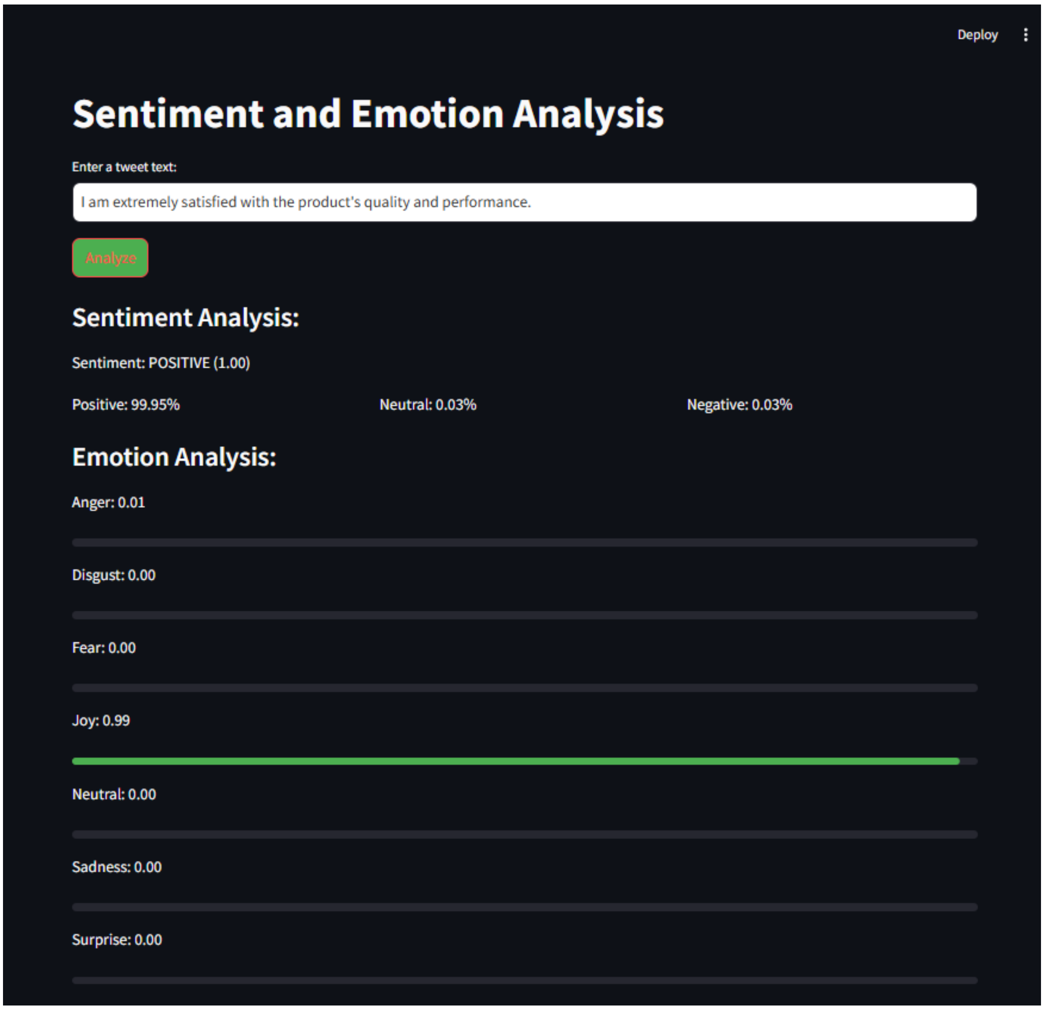

Fig 6.11 - Sentiment and Emotion Analysis Output – Positive Sentiment with High Joy Score

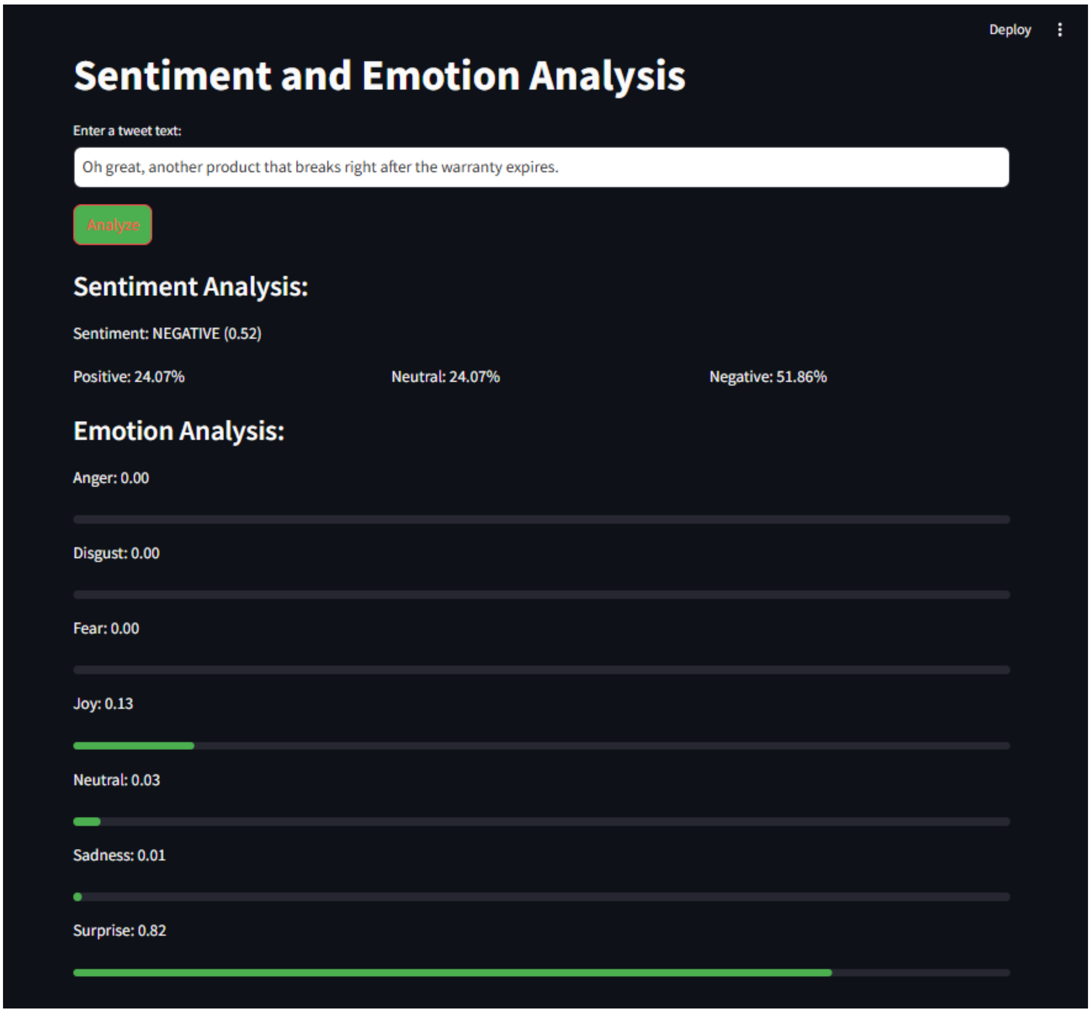

Fig 6.12 - Sentiment and Emotion Analysis Output – Negative Sentiment with Surprise Emotion

2. **Technical Documentation and Reporting**
   - Maintained clean and readable code with proper **commenting, modular functions, and structure**, facilitating easier review and understanding.
   - Documented each project step-by-step using markdown cells, inline explanations, and result interpretations within Jupyter Notebooks.
   - Created a central **GitHub repository** for version control and collaborative access, aligned with modern software engineering workflows.
   - Prepared the report and presentation materials in a structured format suitable for academic submission and professional portfolio use reusable widgets for better maintainability and scalability.

3. **Knowledge Application and Innovation**
   - Applied theoretical concepts such as **data preprocessing, feature engineering, model selection, evaluation metrics, and NLP techniques** to solve real-world problems.
   - Demonstrated the ability to **choose appropriate tools and models** based on task

objectives, resource constraints, and data complexity.

- Explored innovative approaches, such as combining visual insights with model outputs and enhancing dashboards with filter-based interactivity.

4. **Adaptability and Self-Directed Learning**
   - Adapted to multiple environments including **local machines, cloud notebooks (Colab), and visualization tools** without formal training.
   - Leveraged online platforms such as **YouTube tutorials, documentation, forums, and ChatGPT** to troubleshoot issues and accelerate task completion.
   - Demonstrated consistent progress and learning across domains ranging from big data handling to natural language processing.

5. **Professional Growth and Industry Alignment**
   - Aligned all deliverables with real-world use cases, reflecting industry-standard practices in data science and analytics.
   - Gained exposure to full-cycle project execution—starting from problem statement understanding to data preparation, solution development, result interpretation, and final presentation.
   - Delivered outputs in a format suitable for **business stakeholders, academic review, and personal portfolios**, adding significant value to career readiness.

The overall contribution reflects a disciplined, thoughtful, and technically sound approach to project-based learning. The deliverables produced during the internship are not only functional and technically accurate but also exhibit best practices in coding, data visualization, and model interpretability.

CHAPTER 7

REFLECTION AND ANALYSIS

The internship at **CodTech IT Solutions Pvt. Ltd** offered a highly immersive and structured learning experience that bridged academic knowledge with practical implementation. It facilitated a deep understanding of data analytics workflows, machine learning pipelines, and natural language processing techniques through real-world task execution. This chapter provides a critical reflection on the skills developed, knowledge gained, and the overall professional growth resulting from the internship journey.

1. Practical Exposure to Real-World Projects

The internship tasks were designed to reflect real industry scenarios where large datasets, predictive modeling, and text analytics play a central role. Unlike academic projects, which often rely on pre-cleaned datasets and simplified objectives, these tasks required navigating the full data lifecycle—from data acquisition and cleaning to analysis, modeling, and visualization.

Working on unstructured and complex datasets provided the opportunity to understand data variability, quality issues, and the importance of preprocessing. Each task encouraged critical thinking and the application of logic to derive meaningful outcomes, demonstrating how theoretical concepts translate into practical results.

2. Skill Enhancement Across Technical Domains

The internship contributed significantly to skill development in several core areas:

- **Big Data Processing:** Gained confidence in using scalable tools like PySpark to process and analyze large datasets efficiently.
- **Machine Learning Implementation:** Understood the workflow of building predictive models, performing feature engineering, evaluating performance metrics, and deploying classification algorithms.
- **Data Visualization:** Acquired the ability to communicate data-driven insights through dashboards using Tableau and Power BI, emphasizing user-friendly design and interactivity.
- **Natural Language Processing:** Developed knowledge in text analysis, sentiment classification, and the use of NLP frameworks, which are increasingly relevant in today's

AI-driven landscape.

Each task reinforced the need for precision, experimentation, and documentation—key traits in a professional data scientist or analyst.

3. Improved Problem-Solving and Decision-Making Abilities

The internship required addressing several technical challenges independently, including model tuning, data quality issues, and dashboard design decisions. This enhanced the ability to:

- Evaluate multiple solutions for a given problem.
- Select optimal tools and techniques based on task constraints and expected outcomes.
- Interpret complex outputs and translate them into business-friendly insights.

Exposure to such scenarios helped develop structured problem-solving approaches and data-driven decision-making skills.

4. Self-Learning and Adaptability

One of the most valuable outcomes of the internship was the cultivation of **self-learning** habits. The freedom to explore internet-based resources such as documentation, tutorials, and forums instilled the confidence to learn new tools and troubleshoot issues without direct supervision.

This independence mirrored the expectations of modern workplace environments, where adaptability and initiative are highly valued. The experience also underscored the importance of continuous learning, especially in a fast-evolving field like data science.

5. Communication and Documentation Skills

Throughout the internship, emphasis was placed on documenting findings, interpreting model results, and presenting insights through clean code, markdown annotations, and visualization dashboards. These practices significantly improved technical writing and presentation skills.

Deliverables such as:

- Well-commented codebases,
- Insightful notebook narratives,
- Dashboard reports, and

- Structured documentation,

demonstrated the ability to communicate technical outputs effectively to both technical and non-technical audiences—an essential competency in data roles.

6. Alignment with Industry Expectations

The internship tasks closely mirrored real-world applications across sectors such as e-commerce, finance, social media, and customer service. Techniques applied during the internship, such as sentiment analysis and KPI dashboarding, are directly transferable to industry use cases.

This exposure contributed to career readiness by:

- Familiarizing with common tools used in the data industry.
- Enhancing the portfolio with practical, task-based projects.
- Developing the mindset required for analytical roles in a corporate setup.

The internship not only deepened technical expertise but also promoted a holistic understanding of how data can be harnessed to drive strategic insights and innovation. It reinforced the importance of quality, reproducibility, and clarity in data work—elements that are essential for long-term success in the analytics profession.

CHAPTER 8

RECOMMENDATIONS

The internship program at CodTech IT Solutions Pvt. Ltd provided a well-structured and enriching learning experience. However, to enhance the overall effectiveness and ensure even greater value for future participants, several improvements can be considered. One recommendation is to introduce a more detailed orientation session at the beginning of the internship. While the tasks were clearly defined, a formal briefing outlining objectives, deliverables, and expectations could provide interns with a better roadmap for navigating the tasks. This would reduce initial ambiguity and help participants approach the work more strategically.

Another valuable addition could be the inclusion of periodic optional mentorship sessions or check-ins. These interactions, even if brief and virtual, would allow interns to clarify doubts, receive guidance, and share progress with assigned mentors. The current self-paced structure supports independent learning, but the presence of mentor touchpoints would enhance engagement and offer technical support at crucial stages of project development.

In addition, providing access to curated sample projects, annotated code notebooks, and dashboard templates could serve as practical reference materials. These resources would be especially helpful for beginners who are still familiarizing themselves with industry-standard tools. With clear examples to follow, the learning curve can be significantly reduced, allowing interns to focus more on analysis and creativity.

Introducing collaborative tasks as part of the internship could also simulate real-world working conditions more accurately. Collaborative assignments would promote team coordination, effective communication, version control using GitHub, and problem-solving in a shared environment. This approach encourages peer learning and aligns well with professional development goals.

Creating a centralized communication channel or support forum is another enhancement that could prove beneficial. A dedicated platform such as a Slack group or private Discord server would allow interns to exchange ideas, share resources, and seek help in real time. It would foster a sense of community and make the internship experience more interactive and dynamic.
Furthermore, participants should be encouraged to organize and showcase their work in a

professional portfolio or GitHub repository. This would help them build a visible and organized body of work that can be referenced in future job applications. Issuing digital badges or personalized certificates based on task quality and completeness could further validate their achievements.

Finally, the addition of a reflection report or presentation assignment at the conclusion of the internship would be a valuable exercise. It would provide interns with the opportunity to synthesize their learning experience, articulate their approach to problem-solving, and highlight the skills they developed. Such an activity would not only reinforce learning but also prepare them for future professional interactions where presenting work is a key expectation.

Incorporating these suggestions would strengthen the structure, learning outcomes, and overall engagement of the internship program, making it an even more impactful stepping stone for aspiring data professionals.

CHAPTER 9
CONCLUSION

The internship at **CodTech IT Solutions Pvt. Ltd** provided a valuable opportunity to gain practical, hands-on experience in the field of data analysis and machine learning. Through the structured completion of four core tasks—ranging from big data processing and predictive modeling to dashboard creation and sentiment analysis—this internship enabled a holistic understanding of the data science lifecycle. The exposure to real-world datasets and industry-standard tools allowed for the practical application of theoretical concepts, thus bridging the gap between academic learning and professional implementation.

Each task offered a unique learning experience that contributed to skill development in areas such as data preprocessing, scalable data handling, model building, evaluation metrics, data visualization, and natural language processing. The internship not only enhanced technical capabilities but also fostered critical thinking, problem-solving, and adaptability—qualities essential for succeeding in a data-driven environment. Challenges encountered during the course of the internship were met with thoughtful solutions, reinforcing the importance of resilience, experimentation, and continuous improvement.

The emphasis on independent learning and self-paced execution allowed for a deeper exploration of tools like PySpark, scikit-learn, Tableau, and NLP libraries, while also encouraging the use of supplementary online resources for self-guided development. Additionally, the experience of documenting and presenting findings through clean code, dashboards, and structured reports helped cultivate professional communication skills that are highly relevant in industry contexts.

Overall, the internship was a significant milestone in the journey toward becoming a competent data analyst or data scientist. It not only enriched technical knowledge but also instilled confidence in applying analytical thinking to solve complex problems. The program served as a strong foundation for future academic or professional pursuits in the domains of data science, artificial intelligence, and business intelligence. The skills, experience, and insights gained from this internship will undoubtedly prove to be valuable assets in the evolving landscape of technology and data analytics.

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